

Project Design Phase-II
Data Flow Diagram & User Stories

Date	31 Jan 2025
Team ID	LTVIP2026TMIDS36802
Project Name	Empowering India: Analysing The Evaluation of Union Budget Allocation For Sustainable Growth
Maximum Marks	4 marks

Data Flow Diagrams:

1. Raw Dataset

Input Sources:

- Inventory records, production logs, customer orders, supplier databases

Contents:

- Raw data about toy models, materials, production dates, delivery times, defects, pricing, customer feedback, etc.

2. Data Cleaning & Preprocessing

Tools Used:

- Python (data formatting, missing values)
- Excel (initial cleanup and sorting)
- Tableau Prep (merge & filter data from multiple sources)

Processes Include:

- Removing duplicates or errors
- Formatting dates, prices, categories
- Filtering out irrelevant data
- Merging production and customer records for better insights

3. Processed Dataset

State:

- Clean and structured manufacturing data, ready for analysis

Characteristics:

- Organized, reliable, and meaningful
- Attributes like product type, defect rate, region-wise demand, delivery delays

4. Tableau Dashboards

- **Contents:**

- Visuals: Line graphs, bar charts, heat maps, KPIs
- Filters: Toy type, factory location, price range, quality rating
- Maps: Sales distribution, production efficiency by region

- **Purpose:**

- Turn raw manufacturing data into interactive, visual insights for business teams

5. End Users

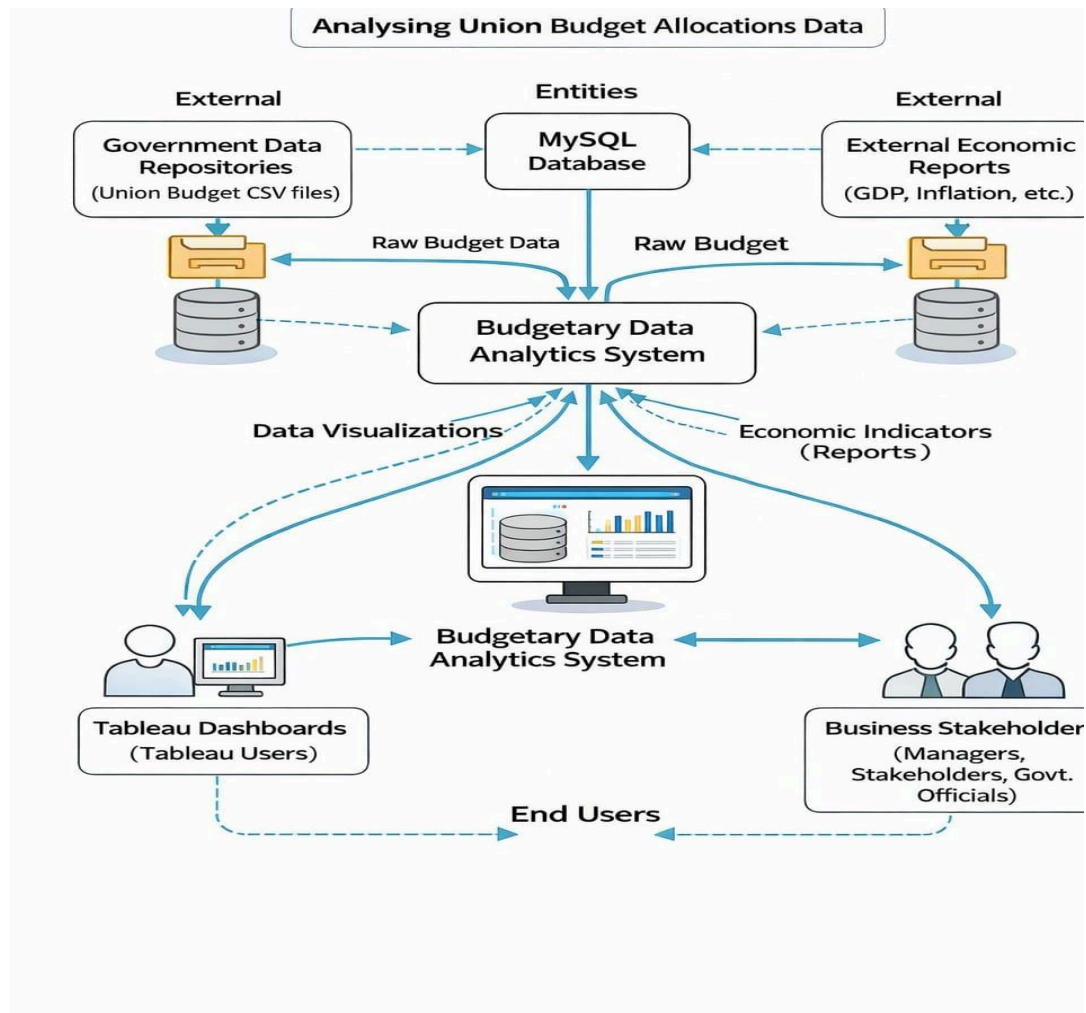
- **Types of Users:**

- Toy Designers – analyze customer preferences
- Production Managers – track delays, improve processes
- Marketing Teams – study sales trends by location
- Distributors – plan inventory based on demand
- Business Analysts – forecast production and profit

- **Interaction:**

- All users explore insights using Tableau dashboards

Example:(Simplified)



User Stories

Use the below template to list all the user stories for the product.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	
Sales Analyst	Sales Performance Monitoring	USN-1	As a Sales Analyst, I can view overall sales performance by product, region, and time period.	I can see total sales revenue, units sold, and average selling price. I can filter data by product category, geographic region, and date range.	High	Sprint-1
		USN-2	As a Sales Analyst, I can analyze sales trends over time to identify growth or decline.	I can view line charts showing sales trends monthly, quarterly, and annually. I can compare current period sales against previous periods.	High	Sprint-1

		USN-3	As a Sales Analyst, I can identify top-performing and underperforming products.	I can view bar charts ranking products by sales volume and revenue. I can drill down into product details to see individual SKU performance.	High	Sprint-2
Inventory Manager	Inventory Optimization	USN-4	As an Inventory Manager, I can view current inventory levels across all warehouses.	I can see stock quantities for each product SKU at each warehouse location. I can identify products with low stock levels.	High	Sprint-1
		USN-5	As an Inventory Manager, I can track inventory turnover rates.	I can view inventory turnover ratios for different product categories. I can	High	Sprint-2

				identify slowmoving or obsolete inventory.		
Production Planner	Production Efficiency	USN-6	As a Production Planner, I can monitor production volumes by product line and factory.	I can view daily, weekly, and monthly production output. I can filter data by specific production lines or manufacturing facilities	High	Sprint-2
		USN-7	As a Production Planner, I can identify bottlenecks or inefficiencies in the production process	I can view data on production cycle times and machine utilization rates. I can compare actual production against planned production targets.	Medium	Sprint-3
Marketing Manager	Marketing Campaign ROI	USN-8	As a Marketing Manager, I can assess the impact of	I can view sales data correlated with specific marketing		

			marketing campaigns on sales.	campaign period I can compare sales before, during, and after a campaign. strategic objectives.	High	Sprint-3
		USN-9	As a Marketing Manager, I can analyze customer demographics associated with product purchases.	I can view customer age groups, locations, and other demographic data linked to popular products.	Medium	Sprint-4
Executive/CEO	Overall Business Performance	USN-10	As an Executive, I can view a high-level overview of key business performance indicators (KPIs).	I can see dashboards with summarized sales, profit margins, inventory health, and customer satisfaction I can quickly identify areas requiring attention.	High	Sprint-1
		USN-11	As an Executive,	I can view variances		

			I can compare actual performance against budget and strategic goals. I can assess progress towards long-term strategic objectives.	High	Sprint-2
Data Administrator	Data Management & Quality	USN-12	As a Data Administrator, I can monitor the data refresh status and data quality. I can view reports on data quality issues or anomalies.	High	Sprint-1
		USN-13	As a Data Administrator, I can manage user access to specific dashboards and data sets. I can view a list of all users and their assigned roles.	Medium	Sprint-2